

SAI ROHITH BHARADWAJ RAMAVARAPU

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EDUCATION

University at Buffalo, SUNY — M.S. Computer Science, Artificial Intelligence
VNR Vignana Jyothi Institute of Engineering & Technology — B.Tech, Computer Science

Expected Dec 2026
May 2024

TECHNICAL SKILLS

Languages: Python, C#, Java, C++, SQL, TypeScript, JavaScript
Backend: ASP.NET Core, FastAPI, REST APIs, Microservices, JWT, PostgreSQL, MySQL, Redis
Cloud & Tools: AWS, Docker, CI/CD, Git, Linux, performance profiling
AI / ML: PyTorch, OpenCV, Scikit-learn, Hugging Face, vector search

EXPERIENCE

University at Buffalo — Graduate Research Assistant, Robotics & Vision Aug 2026 – Present

- Developing adaptive illumination software to improve Visual SLAM robustness in extreme low-light environments.
- Built PyTorch training and evaluation workflows for illumination decomposition and behavior-cloned control policies.

UST — Software Engineer Dec 2024 – Jun 2025

- Built and deployed 12+ C# / ASP.NET Core REST microservices for carbon-emissions analytics across transportation, waste, and supply chains.
- Reduced API response latency by approximately 30% through SQL indexing, connection pooling, complexity improvements, and asynchronous data access.
- Implemented JWT authentication, role-based access control, and structured logging for secure, diagnosable services.
- Delivered Angular components with reactive forms and client-side validation for emissions-data workflows.

Tech Mahindra — Software Engineer Trainee Jan 2024 – Jul 2024

- Built a PDF question-answering pipeline using sentence embeddings, semantic chunking, cosine similarity, and vector search; recorded 0.89 F1 on project evaluation.

South Matrix Institute of Technology — Machine Learning Intern May 2023 – Dec 2023

- Processed and analyzed 500K+ transaction records with Python and Pandas and engineered features for downstream models.

PROJECTS

Carbon Emissions Analytics Platform | .NET, Angular, SQL, Microservices

- Developed service and UI workflows for household, transportation, and waste emissions calculations with authenticated user and administrative experiences.

ShadowFix: Real-World Shadow Removal | Python, PyTorch, OpenCV, Gradio

- Contributed model-comparison evaluation and Gradio workflow testing across RGB, residual, and mask-guided U-Nets on 540 held-out ISTD triplets.
- Analyzed a best result of 30.189 dB PSNR and 0.9514 SSIM, including +3.01 dB overall and +2.21 dB shadow-region PSNR over the RGB baseline.

Video Anomaly Detection | Python, PyTorch, OpenCV, CNN-LSTM

- Implemented a modular video-processing pipeline combining spatial features, motion cues, and temporal sequence modeling for anomaly scoring.

Clinical Risk Prediction API | FastAPI, Scikit-learn, React, Docker

- Benchmarked Random Forest, SVM, and gradient-boosted classifiers, achieved 0.92 ROC-AUC, and exposed the selected model through containerized REST endpoints.

PUBLICATION

“Smart Health Prediction System with Data Mining.” Co-author, *Nanotechnology Perceptions*, Vol. 20, S15, 2024. [Article](#)